Module 2 — Quiz B (Concept Check)

Question bank · After Read 2 · open notes · unlimited attempts

Module 2, Quiz B: Concept check

Reading: Module 2 Read 2 (complete Read 1 + Quiz A first)
Canvas: Concept Check quiz (Module 2)

Apply Read 2: population/sample/parameter/statistic, $\bar{x}/\mu/s/\sigma$, sampling & bias, FPC, z vs t, SE/MOE, mean CI by hand (use 2 as approximate critical value), graphs, and CI interpretation. No R-code items here.

> **!** By-hand multiplier 2 (Quiz A & B)
>
> Use 2 for MOE and CI endpoints on this quiz. Intervals from `t.test()` use t^* and are often wider when n is small — your by-hand answer can be narrower than R. Synthesis tests reading R output.

> **i** Conditions (three layers)
>
> SRS + (BRP for proportions) → large-sample rule (≥ 10 successes and failures for Wald; normal or $n > 30$ for means) → FPC only for finite N with non-tiny n/N. Checklist in Read 2.

R output (`prop.test`, `t.test`), tidyverse, and full Project 2 integration → Synthesis.

Section A — Population, sample, parameter, statistic (Read 2 §0)

A1. Population

Define population in one or two clear sentences.

A2. Sample

Define sample in one or two clear sentences.

A3. Classroom chairs

District $N = 300$ classrooms; SRS $n = 30$; $\bar{x} = 29.8$ chairs.

Identify population, sample, parameter, and statistic.

A4. Course website minutes

25 STAT 109 students; $\bar{x} = 18.4$ minutes/day on course site.

Identify population, sample, parameter, and statistic.

A5. Match symbols

Match each symbol to parameter or statistic:

p, $\hat{p}$, μ, $\bar{x}$, s, σ

Section B — Notation: $\bar{x}$, μ, σ, s, n, N (Read 2)

B1. σ vs s

Which describes population spread (often unknown) vs sample spread from observed data?

B2. μ vs $\bar{x}$

Library hours: you measure $n = 6$ schools; $\bar{x} = 12$ hours.
What is $\bar{x}$? What is μ?

B3. Census

You record library hours for all $N = 120$ colleges in a consortium ($n = N$). Do you need a CI for μ? Why?

B4. Standard error (mean)

Write the formula for $\mathrm{SE}(\bar{x})$ when estimating a population mean from a sample.

Section C — Sampling, bias & frame (Read 2 §1)

C1. Sampling frame

Define sampling frame in one or two clear sentences.

C2. Bias

Define bias in the context of sampling and estimation.

C3. Dining hall intercept

First 60 students outside one dining hall 12–1 p.m. surveyed about sleep.

1. Convenience or SRS? 2. Bias likely? 3. Generalizability?

C4. Registrar SRS

Random-number draw of 120 unique IDs from full undergraduate registrar list.

1. Convenience or SRS? 2. Bias likely? 3. Generalizability?

C5. Club meeting

40 students at pre-med club meeting surveyed about med-school plans.

1. Convenience or SRS? 2. Bias likely? 3. Generalizability?

C6. Incomplete frame

SRS from declared-majors list only; undeclared students excluded.

1. Design label? 2. Bias? 3. Generalizability?

C7. Convenience — generalizability

Researchers survey only students in one dining hall at lunch to estimate sleep for all undergraduates. Why might a correct 95% CI still not generalize?

C8. SRS — carry out

List three ordered steps to draw an SRS of $n = 30$ from a numbered list of $N = 210$ students.

Section D — FPC & method choice (Read 2 §3)

D1. SRS workflow

List three ordered steps for simple random sampling.

D2. Large-state proportion

$n = 220$, $x = 82$ with part-time job; SRS from very large statewide population.

Mean or proportion? FPC? Conditions?

D3. Finite classroom mean

$N = 300$ classrooms, SRS $n = 30$ without replacement; mean chairs.

Mean or proportion? FPC? Conditions?

D4. Clinic proportion with FPC

$N = 80$ patients, $n = 50$ without replacement; Yes/No recovery proportion.

Mean or proportion? FPC? Conditions?

D5. Small count proportion

Large campus SRS; biology major indicator; $\hat{p} = 2/18$.

Mean or proportion? FPC? Conditions?

Section E — z-scores & standardized statistics (Read 2 §2)

E1. Annarose z-score

Women: $\mu = 64$, $\sigma = 2.5$ in. Annarose is 67 in. Compute her z-score.

E2. Compare z-scores

Newton: 72 in; men $\mu = 70$, $\sigma = 3$. Who is more unusually tall relative to their group — Annarose ($z = 1.2$) or Newton?

E3. Standard normal probability

For $Z \sim N(0, 1)$, what does $P(Z \leq 1.24)$ represent? Also find $P(Z > 1.24)$ (table or calculator).

E4. Standardized test statistic pattern

Fill in the pattern: $\dfrac{\text{observed statistic} - \text{reference under } H_0}{\underline{\hspace{2em}}}$

Section F — t vs z, SE, MOE & mean CI by hand (Read 2 §2)

F1. CLT for mean — conditions

$n = 40$ recovery times from a random sample; population shape unknown. Can you use $\bar{x} \pm 2s/\sqrt{n}$ for an approximate 95% CI? Briefly why.

F2. t vs z

When σ is unknown and n is small, why use a t multiplier instead of $z = 2$?

F3. Margin of error

Recovery times: $n = 5$, $\bar{x} = 8$, $s \approx 2.65$, SE ≈ 1.18. Compute MOE using multiplier 2 (by-hand rule).

F4. Mean CI endpoints

Using $\bar{x} = 8$ and MOE ≈ 2.36 from the previous setting, give approximate 95% CI endpoints for μ.

F5. Sample SD by hand

Data: $5, 7, 7, 9, 12$ ($n = 5$). Compute $\bar{x}$ and s (sample SD).

F6. Degrees of freedom

A mean CI uses the t distribution with df $= n - 1$. For $n = 5$ recovery times, what is df?

Section G — Mean, median & graphs (Read 2 §2)

G1. Mean and median

Data: $3, 5, 5, 7, 9$. Compute the mean and median.

G2. Descriptive vs parameter

Recovery times: $\bar{x} = 8$ days, $s \approx 2.65$ from $n = 5$ patients. Which quantities are statistics describing this sample?

G3. Dot plot vs histogram

For $n = 8$ integer quiz scores, which plot shows each duplicate value clearly?

G4. Histogram idea

What does a histogram show for one numerical variable?

Section H — Interpreting CIs in context (Read 2 §0)

H1. Interpret proportion CI

95% CI for proportion of students who ate breakfast today: [0.42, 0.58]. Write one sentence using *I am 95% confident that...*

H2. Interpret mean CI

95% CI for mean weekly library open hours: [18.2, 24.6] hours. Write one sentence using the interpretation recipe.

H3. Mean CI conditions checklist

Before trusting a mean CI, name two checks from the conditions checklist (sampling + model).

Section I — Variable types (methods map)

I1. Variable types

Classify each as numerical or categorical (and binary if categorical):

a) Stem diameter (cm)

b) Soil type: sand/loam/clay

c) Parasite present: yes/no

d) Egg count per nest
